

Evaluation of Delfino (TMS320F28379D) Processor for Helmet ANC Application

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Abstract— Embedding Active Noise Control (ANC) system to fighter aircraft helmet imposes size constraint for the controller board as it needs to be integrated inside ear cup. Further ANC system development needs to take place under stringent weight and power constraints. Selection of a suitable processing platform plays a key role in meeting the requirements. This paper proposes to use TMS320F28379D, a microcontroller with DSP capabilities. Evaluation is done by carrying out single channel ANC simulation. Also, the size, weight and power consumption of the ANC controller achievable with this processor is worked out to ascertain that it can be integrated into the helmet.

Keywords—Active Noise Control, FXLMS Algorithms, Adaptive filters, Pilot helmet, Feed forward ANC, ANC Causality Condition, TMS320F28379D

I. INTRODUCTION

The noise environment in fighter aircraft cockpit can reach the levels as high as 105dBA[1]. Since protection is of paramount importance for a fighter aircraft pilot, he wears a certified helmet made of high strength, lightweight material. In addition to protection, the helmet attenuates most of the high frequency noise components, accounting to about 15-20dBA passive attenuation [1]. In spite of this attenuation, prolonged exposure to noise levels greater than 90dBA can lead to hearing impairment and may affect his judgment capability. Further, speech and recognition of warning signals is affected by the noisy environment which leads to masking of voice communication signals. Hence noise levels need to be reduced even further and since the contribution inside the helmet is mainly due to low frequency, ANC technology can be effectively used [2].

Anti noise generating loudspeakers, microphones and a processor for running ANC software are the major components which have to be integrated to the helmet. Typically helmet ANC systems are battery powered as routing of aircraft power to helmet requires modifications in aircraft. Since helmet ANC system is a pilot wearable equipment, the weight of the ANC system has to be very light as weight experienced varies with changing 'g' values. The major component which decides the system weight is the battery. Hence power consumption of the ANC components must be optimized to bring the weight down to acceptable value. Seamless integration of ANC component to

helmet is essential and that calls for compactness of the controller.

Hence there is a necessity for identifying a suitable processing platform. Since ANC algorithms are based on adaptive signal processing, they are computationally intensive and DSPs have been the obvious choice [3]. In automotive industries even PowerPCs have been employed [11]. However, their power consumption is quite high. ANC has also been implemented on microcontrollers in head phone applications [4], but may not be suitable to helmet ANC application as additional coding has to be incorporated to meet the aircraft safety requirements [5].

In view of this, a microcontroller with DSP capability is an attractive option and this paper proposes to assess the feasibility of Texas Instruments' Delfino series TMS320F28379D microcontroller for helmet ANC application.

Section II gives details of helmet ANC system and its requirements. Section - III describes the features of TMS320F28379D microcontroller from ANC perspective. Section IV gives the evaluation of the microcontroller through simulation.

II. HELMET ANC SYSTEM

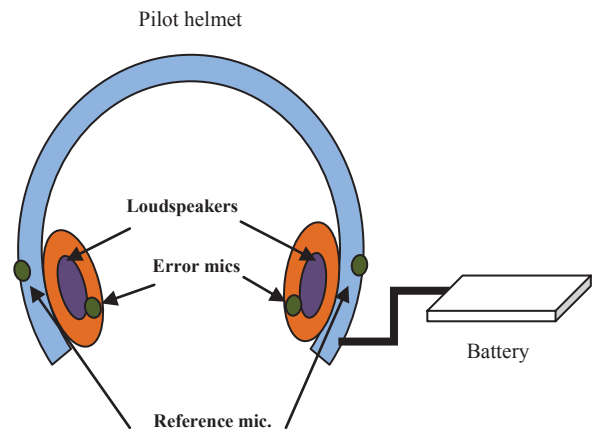


Fig. 1. Block Diagram of Helmet ANC

ANC is an electro acoustic system that cancels the primary noise by generating the anti noise signal of equal

amplitude with opposite phase [3]. Feed forward ANC is the preferred scheme as it ensures pilot communication [3]. The placement of ANC transducers inside the helmet earcups is as shown in Fig 1. The ANC controller board is integrated inside the ear cup of the helmet and battery is connected externally. The ANC controller board comprises of the processing element and signal conditioning elements for loudspeakers and microphones.

A. Causality condition leading to sampling

The anti noise generation rate should meet the causality condition, that is electrical delay (computed from reference microphone pickup to anti noise coming out of loudspeaker) should be less than acoustic delay (time taken by acoustic signal to travel from reference microphone position to loudspeaker position.). Acoustical delay translates to physical distance between reference microphone and loudspeaker. In case of helmet ANC this distance is as small as 4-5 cm.

The acoustical delay is quantified as [3],

$$\delta_a = \frac{d}{S} \quad (1)$$

where S = Speed of sound in air, 330m/s
 d = distance in meter

The electrical delay is quantified as [3],

$$\delta_e = T \left(1 + \frac{3}{8} (2M) \right) \quad (2)$$

where T is the processor sampling time
 M is the order of anti aliasing/reconstruction filter

TABLE I. ACOUSTICAL AND ELECTRICAL DELAYS IN HELMET ANC

Parameters	Value
Distance between Ref Mic and Loudspeaker (m)	0.05
Acoustical delay(s)	0.00015
Processor Sampling Frequency (KHz)	38

ANC application requires sharp cutoff and hence 8th order filter is used. With this, to meet this constraint the ANC controller should operate at minimum of 38KHz (Table I).

B. ANC algorithm

Helmet ANC requires a single channel ANC for each ear cup. For real-time application, FXLMS algorithm is best suited in terms of its simplicity and minimal computational complexity.

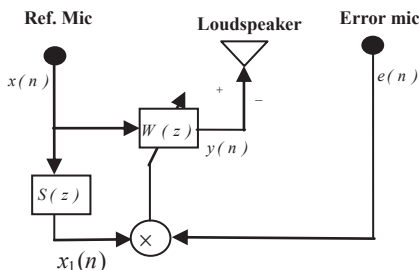


Fig. 2. Block diagram of ANC algorithm

The Feed forward FXLMS algorithm is as follows [3]. The reference microphone pickup $x(n)$ is filtered with adaptive filter $W(z)$ to obtain anti noise $y(n)$, which is driven out through the loudspeaker. The error microphone pickup $e(n)$ is used for adaptation of $W(z)$. FXLMS algorithm uses filtered version of the input to account for the delays associated with the secondary path [6]. The equations used are as follows:

$$y(n) = \sum_{i=0}^{L-1} w_i(n) b_i(n) \quad (3)$$

$$x_1(n) = \sum_{m=0}^{M-1} s^m(n) x(n-m) \quad (4)$$

$$w_i(n+1) = w_i(n) + \frac{\mu}{P_{ze}} x_1(n) e(n) \quad (5)$$

$$P_{ze1} = \beta(P_{ze1}) + (1-\beta) x_1^2 \quad (6)$$

C. Computation complexity

The computational complexity of FXLMS algorithm is computed using equations (3)-(6), in terms of number of multiply-Accumulate (MAC) operations. Table II gives the computations and memory requirements involved for a single channel ANC.

TABLE II. COMPLEXITY AND MEMORY REQUIREMENTS OF SINGLE CHANNEL FXLMS ALGORITHM

Operation	Number of MACs	Memory requirements
Main path filtering	L	L
Filtered input	M	M
Coefficients +power adaptation	L+1	L+1
Total	2L+M+1	2L+M+1

III. TMS320F28379D MICROCONTROLLER FOR ANC

TMS320F28379D is 32 bit microcontroller with DSP capability. It has dual core C28x DSP processor each core operating at 200MHz and 2 CLA operating at 200MHz concurrent with processor to accelerate math operations[8].

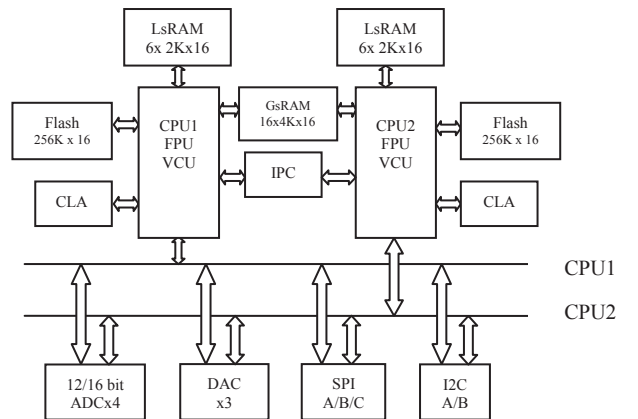


Fig. 3. Architecture of TMS320F28379D MCU

Each of the C28x CPU core are designed around 32-bit fixed point accumulator architecture, which is mix of both DSP and microcontroller architecture, providing excellent 32-bit processing capabilities. From pure architecture point of view it uses a modified Harvard architecture, which

enables instructions and data reads to be performed in parallel, along with data writes. The processor comes with six separate address/data bus and with 8-stage pipeline.[6,7]

Addition of FPU to the C28x fixed point CPU core supports IEEE single-precision floating point operations. The Viterbi, Complex Math and CRC unit (VCU) is very useful for general-purpose signal processing.[7,8]

Further each C28x core has a Control Law Accelerator (CLA), which is an independent 32-bit floating-point processor, which runs concurrently with the same speed of the main CPU. CLA can boost the total performance of the device to 800 MIPS[7] with each CPU running at 200MHz, which helps in achieving the higher sampling rates. The Inter Processor Communication (IPC) module helps in communication between the two CPU subsystems with two dedicated 1Kx16 blocks of message RAM for transfer of message and data between CPU1 and CPU2.

There are four independent 16/12-bit ADC modules with sampling frequency 1.1MSPS/3.5MSPS respectively to interface with microphones. The microcontroller also has 3 buffered 12-bit DACs(settling time of 2 μ s) to drive loudspeakers through power amplifiers. The microcontroller has inbuilt flash of 1MB for storing the application program.

A. TMS320F28379D power requirement

The power consumption of MCU is proportional to the system clock frequency(MHz). At 200MHz system clock frequency, TMS320F28379D consumes 820mW [12].

IV. EVALUATION OF TMS320D28379D FOR HELMET ANC

A. I/O requirement

Helmet ANC application requires 3 ADCs and 2 DACs. From I/O resource point of view, it can be concluded that the resources available on the chip are sufficient for ANC application.

B. ANC computation assessment

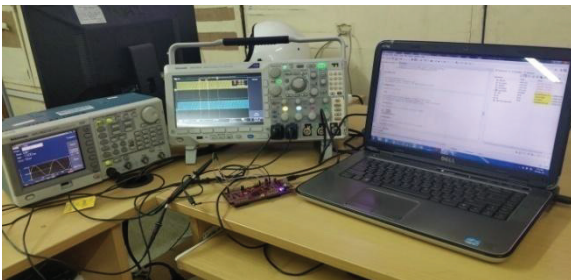


Fig. 4. ANC evaluation setup

The experimental setup consists of Function Generator, Mixed Digital oscilloscope, TMS320F28379D MCU Launch Pad with JTAG emulator connected to laptop (Fig. 4). To evaluate the performance, a single channel ANC algorithm was implemented in C language. A 500Hz sine wave generated by signal generator served as the primary noise and is fed to ADC. Simulated main path and secondary path coefficients as given in [9] were used for

desired signal generation. The error signal was fed to DAC to observe the convergence on the oscilloscope.

Single channel ANC cycle consumption was analyzed with all the optimization levels offered by the CCS compiler. Table III shows that Level 2 Global Optimization takes 8610 cycles and this amounts to sampling rate of 23KHz. Even with this, the causality condition is not satisfied.

TABLE III. SIMULATION RESULTS OF SINGLE CHANNEL ANC

Optimization levels	Number of Clock Cycles	Sampling Rate in KHz
Off	12140	16
Register-0	10653	18
Local-1	10395	19
Global-2	8610	23

FXLMS algorithm is characterized by Multiply and Accumulate (MAC) instructions executed in a loop. Hence, the loop unrolling technique can be employed to reduce the loop overheads. Loop unrolling (manual) technique is also called space-time tradeoff approach which leads to increase in the program execution speed at the expense of program size [9]. Since in case of ANC, code size is about 15KB, which is very much less than the memory capacity of TMS320F28379D MCU, loop unrolling can be considered.

In loop unrolling, four MAC statements were executed per loop instead of single MAC statement. This reduces loop overhead by 75%. With Level 2 Global Optimization and loop unrolling, a sampling rate of 48.5KHz was achieved, thus meeting the causality criterion.

Thus, from computation perspective TMS320F28379D can be considered for development of ANC software.

C. ANC Controller size assessment

In a typical helmet ear cup, space available for integration of ANC controller board is about **30mm X 60mm**. ANC controller board comprises of transducer signal conditioner electronics apart and processing hardware. The electronic components comprise of power amplifiers, pre amplifiers and filters which consumes about 60% of the board size. The TMS320F28379D MCU is of **16mm X 16mm**, which is about 15% of board size and hence can be easily accommodated in the 40% space available.

D. ANC Controller power consumption assessment

Power consumption by the ANC controller based on TMS320F28379D can be assessed as in Table IV.

TABLE IV. POWER CONSUMPTION OF ANC COMPONENTS

Sl No	Components	Wattage in mW
1	TMS320F28379D	820
2	Power amplifiers	200
3	Other electronics	100
	Total	1120

It is required that the ANC system should be operational continuously for about 4 hours. For this, the battery requirement is worked out to be a Lithium Polymer battery of 2200mAh at 3.7V.

E. ANC Controller weight assessment

The weight of the controller assessment is done with 2 PCBs embedded into ear cups, the battery with enclosure and associated cable with connectors. The weight of the loudspeakers are not considered as the loudspeakers present in the ear cup as a part of aircraft communication will be used for ANC.

TABLE V. WEIGHT OF ANC COMPONENTS

Sl No	Components	Weight in grams
1	Controller PCBs	30
2	Battery with enclosure	70
3	Cable and connectors	40
4	Microphones	10
	Total	150

V. CONCLUSION

The Delfino MCU TMS320F28379D was evaluated for helmet ANC application in terms of meeting computational load, size, weight and power requirements. The single channel ANC code implemented in single C28x processor core with global optimization and loop unrolling techniques satisfied the causality condition of 38KHz. Further, since the processor has two CPUs, both ANC channels can be accommodated, thus realizing helmet ANC in a single chip. Also, size of the Delfino MCU based controller board is such that it can be easily embedded into the ear cup. The weight of the controller unit is about 150g.

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